

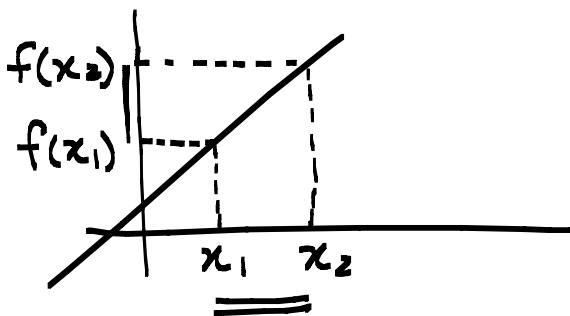
Unit 4.3 How derivatives affect the shape of the graph. p 290 - 301

Note Title

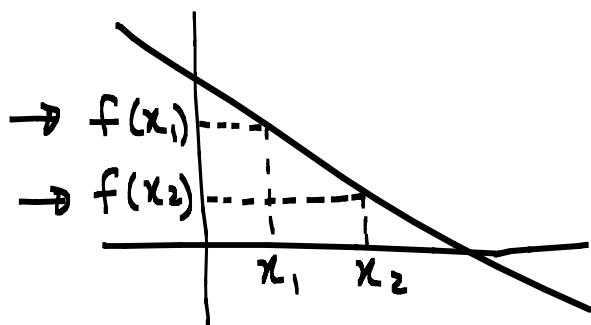
2014/04/07

Remember p 19:

f is increasing on interval I if
 $f(x_1) < f(x_2)$ whenever $x_1 < x_2$ in I .



f is decreasing on interval I if
 $f(x_1) > f(x_2)$ whenever $x_1 < x_2$ in I .



Increasing/Decreasing Test p 290

- (a) If $f'(x) > 0$ on an interval, then f is increasing on the interval.
- (b) If $f'(x) < 0$ on an interval^I, then f is decreasing on the interval.

Proof:

(b) let x_1 and x_2 be any two numbers in the interval with $x_1 < x_2$.

We must prove that $f(x_1) > f(x_2)$.

Given that $f'(x) < 0$:

f is differentiable on $[x_1, x_2]$ and by the MVT we have a number

$c \in (x_1, x_2)$ so that

$$f'(c) = \frac{f(x_2) - f(x_1)}{x_2 - x_1}$$

$$\Rightarrow f(x_2) - f(x_1) = f'(c)(x_2 - x_1)$$

Now: $f'(c) < 0$ (given) and $x_2 - x_1 > 0$

$$\Rightarrow f(x_2) - f(x_1) < 0$$

$$\Rightarrow f(x_2) < f(x_1) \quad \blacksquare$$

Examples:

① Find the interval(s) on which $f(x) = xe^x$ is decreasing.

$$f'(x) = e^x + xe^x = e^x(1+x)$$

f will decrease if $f'(x) < 0$

$$\Rightarrow e^x(1+x) < 0$$

$$\Rightarrow (1+x) < 0 \quad \text{since } e^x > 0 \quad \forall x$$

$$\Rightarrow x < -1$$

So f decreases on $(-\infty, -1)$

(2) Determine the interval(s) on which $f(x) = e^{2x} - 6x$ is increasing.

$$f'(x) = 2e^{2x} - 6$$

f increases if $f'(x) > 0$

$$\Rightarrow 2e^{2x} - 6 > 0$$

$$\Rightarrow e^{2x} > 3$$

$$\Rightarrow 2x > \ln 3$$

$$\Rightarrow x > \frac{1}{2} \ln 3$$

$\Rightarrow f$ increases on $(\frac{1}{2} \ln 3, \infty)$

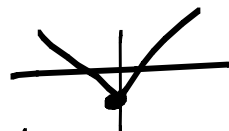
The First Derivative Test p 291

c is a critical number of continuous function f :

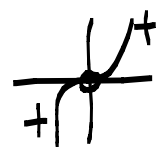


$$f'(c) = 0$$
$$f'(c) \neq$$

(a) f' changes from $+$ to $-$ at c , then f has a local max at $x = c$.



(b) f' changes from $-$ to $+$ at c , then f has a local min at $x = c$.



(c) f' does not change sign at c , then there is no local max/min at c .

Examples:

① Find the local extremes of $f(x) = -x^5 + 15x^3$

f is a polynomial $\Rightarrow f$ is cont on $\mathbb{R}$

$$f'(x) = -5x^4 + 45x^2$$

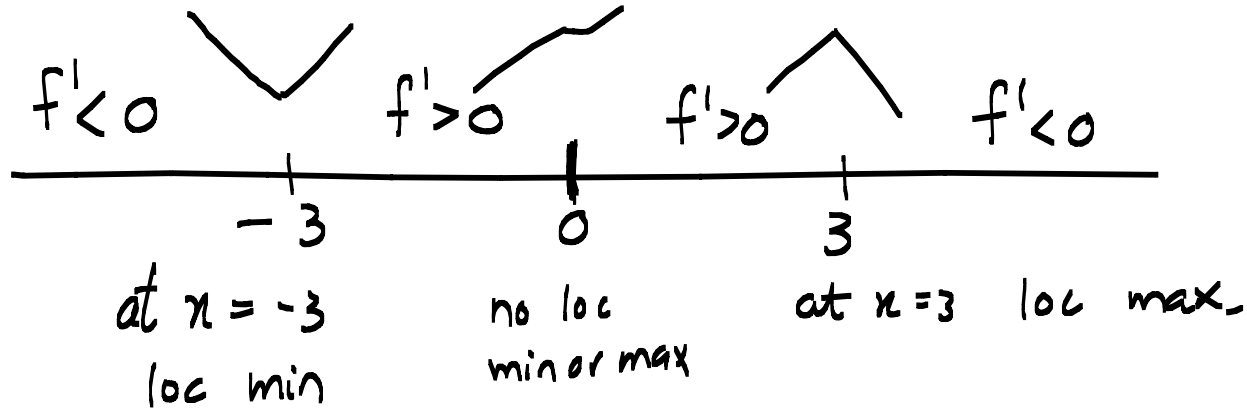
f' is continuous everywhere. $f'(x)$ exists everywhere

$$f'(x) = -5x^4 + 45x^2 = 0$$

$$\text{So } \underbrace{-5x^2}_{+} (\underbrace{x-3}_{+}) (\underbrace{x+3}_{+}) = 0$$

Critical numbers at

$$x = 0, \quad x = 3, \quad x = -3$$



loc min of $f(-3) = -162$

loc max of $f(3) = 162$

f' does not change sign at $x=0 \Rightarrow$ no local extreme.

- ② Find the local min/max of the given function if they exist.

$$f(x) = \frac{x}{x^2 + 4}$$

$$f'(x) = \frac{4 - x^2}{(x^2 + 4)^2} \text{ which exists on } \mathbb{R}.$$

$$\text{let } \frac{4 - x^2}{(x^2 + 4)^2} = 0$$

$$\Rightarrow x = \pm 2$$

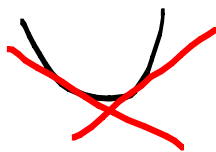
Use a number line!! $\Rightarrow$ Critical numbers at $x = -2$ and 2

A number line with points -2 and 2 marked. Above the line, the sign of f' is indicated: $f' < 0$ for $x < -2$, $f' > 0$ for $-2 < x < 2$, and $f' < 0$ for $x > 2$.

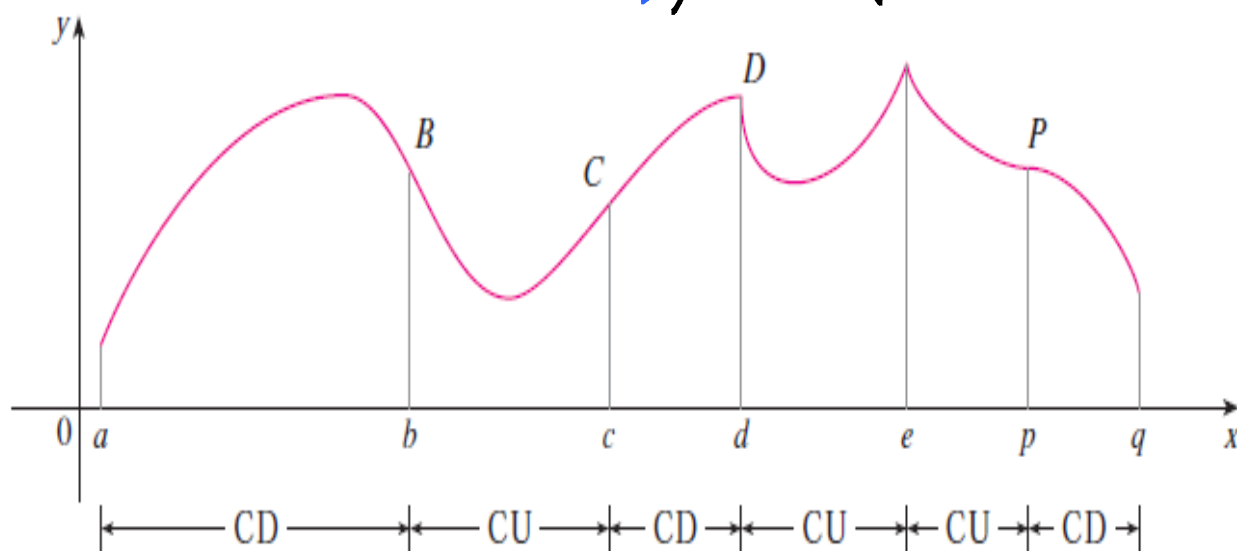
loc min $f(-2) = -\frac{1}{4}$ and loc max $f(2) = \frac{1}{4}$

Definition p 293

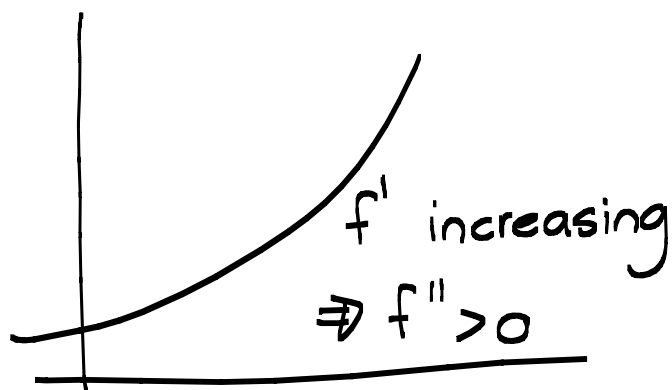
If the graph of f lies above all its tangent lines: Concave upward on I



If the graph of f lies below all its tangent lines: Concave downward

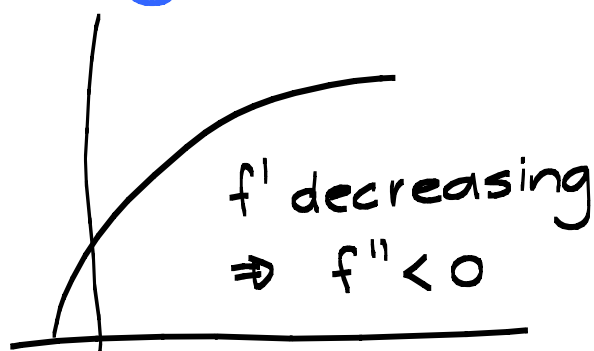


How do we test concavity?



$$f''(x) > 0$$

f concave upward



$$f''(x) < 0$$

f concave downward.

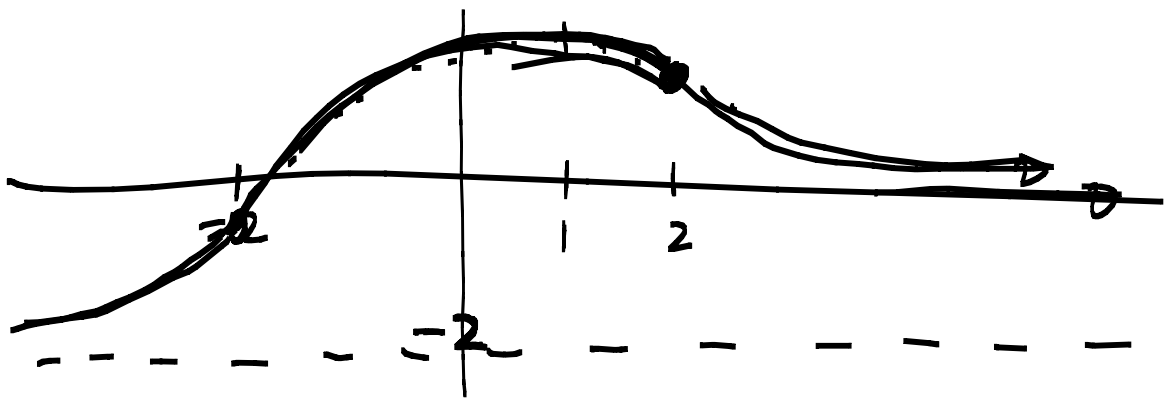
Examples:

Sketch a graph satisfying the following

(i) $f'(x) > 0$ on $(-\infty, 1)$; $f'(x) < 0$ on $(1, \infty)$

(ii) $f''(x) > 0$ on $(-\infty, -2)$; and $(2, +\infty)$
 $f''(x) < 0$ on $(-2, 2)$

(iii) $\lim_{x \rightarrow -\infty} f(x) = -2$, $\lim_{x \rightarrow \infty} f(x) = 0$



The Second Derivative Test p 295

f'' is continuous near c .

(a) If $f'(c) = 0$ and $f''(c) > 0$ then f has a local min at c .

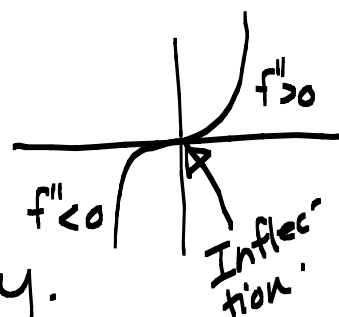
(b) If $f'(c) = 0$ and $f''(c) < 0$ then f has a local max at c .

If $f''(c) = 0$ the test fails.

Inflection point p 294

Where the graph changes concavity.

From concave upwards to downwards or from concave downwards to upwards.



Examples:

Find the local extremes using the 2nd derivative test.

1. $f(x) = -x e^{-3x}$

$$f'(x) = -e^{-3x} + 3x e^{-3x}$$

$$f'(x) \text{ exists on } \mathbb{R}$$

$$\text{Let } -e^{-3x} + 3x e^{-3x} = 0$$

$$\rightarrow e^{-3x}(-1+3x)=0$$

$$\rightarrow -1+3x=0 \quad \text{since } e^{-3x} \neq 0$$

$$\Rightarrow x = \frac{1}{3} \text{ is the only}$$

critical point.

$$f'(x) = e^{-3x}(-1+3x)$$

$$f''(x) = 3e^{-3x}(2-3x)$$

$$f''\left(\frac{1}{3}\right) = 3e^{-1}(2-1) > 0$$

$\Rightarrow$ at $x = \frac{1}{3}$ we have a
local min of $f\left(\frac{1}{3}\right) = -\left(\frac{1}{3}\right)e^{-1}$

2. Find the local extremes using the
2nd derivative test if

$$f(x) = \frac{\ln x}{x}$$

$$f'(x) = \frac{1 - \ln x}{x^2}$$

$$f''(x) = \frac{2 \ln x - 3}{x^3}$$

$f'(x)$ does
not exist
if $x = 0$

To find the critical points let

$$\frac{1 - \ln x}{x^2} = 0$$

$$\text{So } x \neq 0 \text{ and } \ln x = 1$$

$$x = e$$

Critical points at $x=0$ and $x=e$

$f''(0)$ is not defined.

$$f''(e) = \frac{2-3}{e^3} < 0$$

$\Rightarrow f$ is concave down at $x=e$

$\rightarrow f$ has a loc. max at $x=e$
which is $\underline{f(e)} = \frac{\ln e}{e} = \bar{e}$

(2.) Find the inflection points of

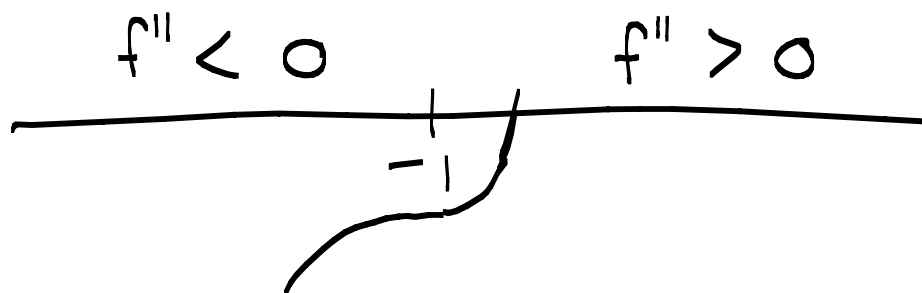
$$f(x) = 1 + 3x + 3x^2 + x^3$$

$$f'(x) = 3 + 6x + 3x^2$$

$$f''(x) = 6 + 6x$$

$$\text{let } 6 + 6x = 0$$

$$x = -1$$



$\Rightarrow f$ has an inflection point at $x = -1$

(A) (a) Find the critical points of
 $f(x) = x^4(x-1)^3$

(b) What does the 2nd derivative test say about these?

(c) What does the 1st derivative test say about these?

(a) $f'(x) = 0$ for critical points!

$$f'(x) = 4x^3(x-1)^3 + x^4 \cdot 3(x-1)^2$$
$$= \frac{x^3(x-1)^2 [4(x-1) + 3x]}{\quad} \quad (-1)$$

$$x = 0, 1 \text{ and } \frac{4}{7}$$

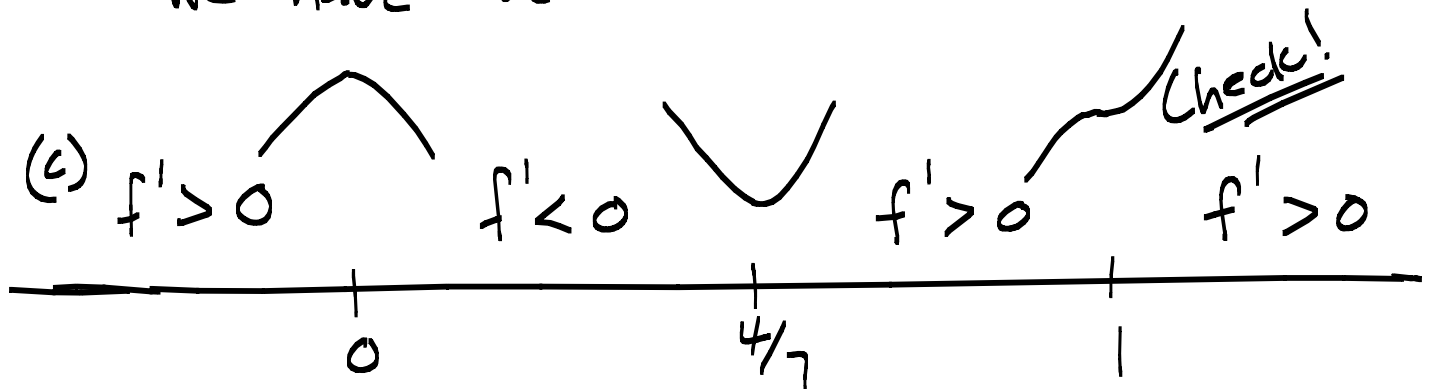
$(fgh)' = f'gh + fg'h + fgh'$

(b) $f''(x)$

$$= 3x^2(x-1)^2 [7x-4] + x^3 \cdot 2(x-1) [7x-4]$$
$$+ x^3(x-1)^2 \cdot 7$$

$$\left. \begin{aligned} f''(0) &= 0 \\ f''(1) &= 0 \\ f''\left(\frac{4}{7}\right) &> 0 \end{aligned} \right\} \text{Check!}$$

Test fails for $x=0$ and $x=1$ and
we have a loc min $f\left(\frac{4}{7}\right)$ at $x = \frac{4}{7}$



loc max $f(0)$ at $x=0$

loc min $f\left(\frac{4}{7}\right)$ at $x = \frac{4}{7}$

$x=1$ is neither a loc max/min.

- ⑤ Discuss the curve $y = x^4 - 4x^3 + 3$
w.r.t. concavity, points of inflection
local max/min.

Use this information to sketch the graph

$$y' = 4x^3 - 12x^2$$

$$y'' = 12x^2 - 24x$$

- ① To find the critical points let

$$4x^3 - 12x^2 = 0$$

$$\Rightarrow x^2(x-3) = 0$$

$\Rightarrow$ Critical points at $x=0$ and $x=3$

- ② The 1st derivative test

$$\begin{array}{c} f' < 0 \qquad f' < 0 \qquad f' > 0 \\ \hline \qquad \qquad 0 \qquad \qquad 3 \end{array}$$

There is a local min at $x=3$
namely $f(3) = -24$

- ③ To discuss concavity let

$$12x^2 - 24x = 0$$

$$\Rightarrow x = 0 \quad \text{or} \quad x = 2$$

$$\begin{array}{c} f'' > 0 \qquad f'' < 0 \qquad f'' > 0 \\ \hline \qquad \qquad 0 \qquad \qquad 2 \end{array}$$

Inflection points at $x=0$ and $x=2$

